

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

COMPUTER SCIENCE
PAPER 2 Structured

4021/2
2 hours

SPECIMEN PAPER

Additional materials:
Scientific calculator [non-programmable]

This booklet should not be punched or stapled and pages should not be removed.

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number at the top of every page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question.

This specimen paper consists of 13 printed pages.

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1 (a) Give **two** examples of print outs produced by the graphic plotter.

- 1 _____
- 2 _____ [2]

(b) (i) State **two** utility softwares that deal with the security of the computer.

- 1 _____
- 2 _____ [2]

(ii) Explain the utilities you mentioned in **b(i)**.

- Utility 1 _____
- _____
- _____
- _____
- Utility 2 _____
- _____
- _____
- _____
- _____ [4]

- (c) Explain **two** ways a computer technician can use to solve the following hardware and software problems.

- (i) RAM faults and errors.

1. _____

2. _____

[4]

- (ii) Corrupted drivers

1. _____

2. _____

[4]

2 (a) Explain **two** importance of computers in the following areas.

(i) Mining

1. _____

2. _____

_____ [2]

(ii) Wildlife Management

1. _____

2. _____

_____ [2]

(iii) Research

1. _____

2. _____

_____ [2]

(b) (i) Define the term *intelligent system*.

_____ [2]

(ii) Explain **two** disadvantages of intelligent systems.

1. _____

2. _____

_____ [2]

3 (a) Showing all working, convert the binary number 111010 to:

(i) Base 10

[2]

(ii) Base 8

[2]

4 (a) Describe the following types of networks.

(i) LAN (Local Area Network)

[2]

(ii) WAN (Wide Area Network)

[2]

(b) (i) Draw a star Topology. [4]

(c) Differentiate between

(i)

HUB

SWITCH

[3]

(ii)

BRIDGE

ROUTER

[3]

5 (a) Explain the term *back up*.

[2]

(b) List **three** methods a bank can adopt to back up its data.

1.

 2.

 3.

- [3]

(c) Outline **three** measures a school can put in place to protect its data from theft and corruption.

1.

 2.

 3.

- [3]

(d) Explain the term *hacking*.

[2]

- 6 (a) Describe **three** activities that take place in the design stage of SDLC.

1. _____

2. _____

3. _____
_____ [6]

- (b) Identify **two** data collection techniques used in SDLC.

1. _____

2. _____
_____ [2]

- (c) Explain the term *technical documentation*.

_____ [2]

- 7 A teacher has 5 students who wrote a test marked out of 25. Using the top-down approach, write a pseudo code which process each mark and output it as a percentage.

Hint: Input and validate each mark.

Calculate the percentage for each mark.

Display the mark out of 25 and the percentage mark.

[10]

8 (a) List **three** runtime errors.

1. _____

 2. _____

 3. _____

- [3]

(b) Describe the term *testing* in a programming context.

[2]

9 Write a Structured Query Language (SQL) command that show all Learners in the Students Table who are taught by Madhara whose major course is computer science.

[6]

10 (a) List **three** webpage elements.

1. _____
2. _____
3. _____ [3]

(b) State **two** functions of web navigation.

1. _____
2. _____ [2]

11 (a) Define the following as they relate to intellectual property rights.

(i) Patents

 [2]

(ii) Copyright

 [2]

(b) Describe **two** elements of intellectual capital.

1. _____

_____ [2]

2. _____

_____ [2]

(c) Explain why attributes of business ethics are important in an organisation.

_____ [2]